

PRACTICAL REPORT

COURSE – BUSINESS ANALYTICS (UNIT 2)

PRACTICAL 6

**BUILD THE RETAILPULSE CUSTOMER INSIGHTS
DASHBOARD USING IBM COGNOS ANALYTICS**

SUBMITTED BY

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1. AIM / OBJECTIVE

To assemble a dashboard that combines a summary card, a trend chart, a segment breakdown and a shared filter widget for RetailPulse customer data. Related topic: Build a Dashboard for Customer Insights Visualization.

2. MATERIALS REQUIRED

- IBM Cognos Analytics access.
- RetailPulse Customers data module.

3. INTRODUCTION

A **dashboard** in IBM Cognos Analytics brings several visualizations together on one canvas so that users can monitor key figures and explore patterns at a glance. Each visualization is a **widget** connected to a data source, here the **RetailPulse Customers data module**, which exposes business-friendly items such as Region, Loyalty Segment and customer counts.

The dashboard in this practical uses four kinds of widget. A **summary card** shows one headline figure, here total active customers. A **line chart** shows how values change over time, here new versus returning customers. A **bar chart** compares categories, here customers by loyalty segment. A **filter widget** lets the viewer choose a Region, and because the filter is shared, **cross-filtering** updates every widget on the canvas at the same time.

A good dashboard also follows a **visual hierarchy**: the most important figure and the controls are placed prominently at the top, the main trend gets the largest area, and supporting breakdowns sit beside or below it without overlapping.

4. PROCEDURE

Step	Action
1	Create a new Dashboard and connect it to the RetailPulse Customers data module as its source.
2	Add a summary card widget showing total active customers for the selected period.
3	Add a line chart with a date item on the horizontal axis and the customer count split into New and Returning customers.
4	Add a bar chart showing the customer count for each loyalty segment.
5	Add a Region filter widget, set it to apply to all widgets, then select regions and confirm that all three widgets change.
6	Arrange the widgets into a clear visual hierarchy with no overlaps, then save the dashboard with a clear name.

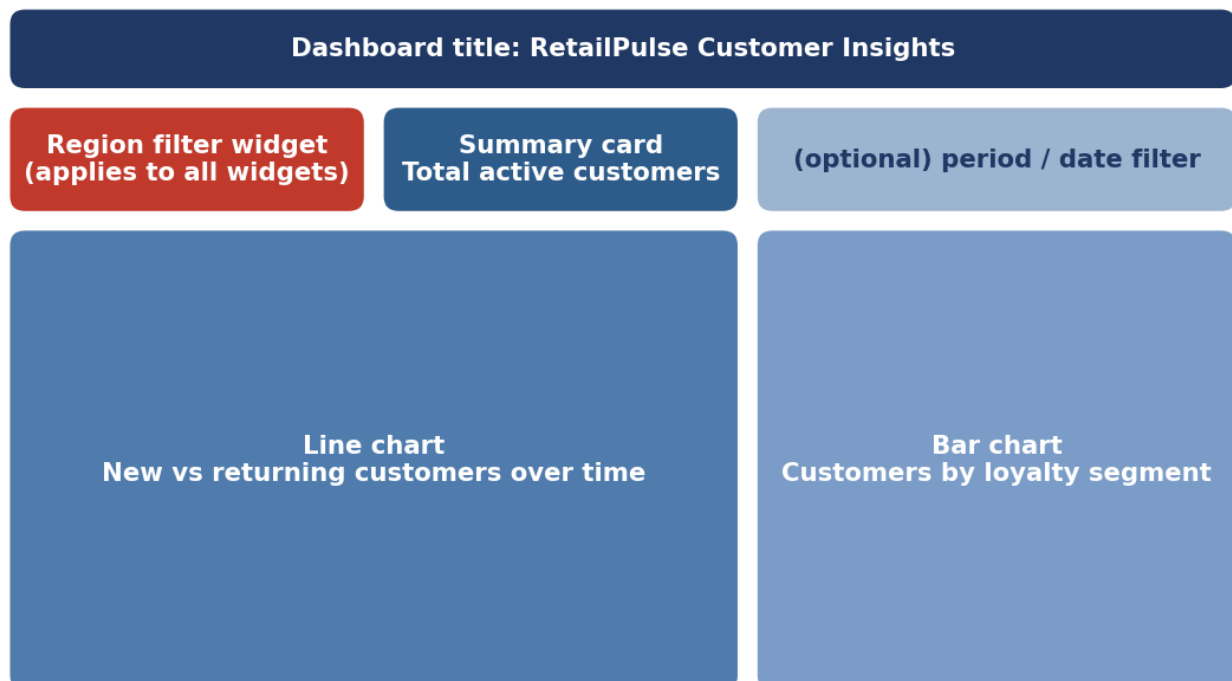
5. WIDGET CONFIGURATION

Item names below are examples. Use the equivalent items in your data module.

Widget	Data items (example)	Purpose
Summary card	Customer count measure (active customers)	Headline figure for the selected period
Line chart	Date (month) on the x-axis; customer count; Customer Type (New / Returning) as series	Trend of new versus returning customers
Bar chart	Loyalty Segment as category; customer count as value	Customer mix by loyalty segment
Filter widget	Region	Shared filter for all widgets

6. PLANNED LAYOUT

Planned layout (schematic, not real output)



The summary card and filter sit at the top because they frame everything below. The trend line gets the widest area, and the segment bar chart sits beside it.

7. OUTPUT SCREENSHOTS

Insert screenshots from your own dashboard in the boxes below.

Screenshot 1: complete dashboard with all widgets and no region selected

Screenshot 2: widget configuration (data items for the line chart or summary card)

Screenshot 3: dashboard after selecting a region in the filter widget

Screenshot 4: saved dashboard in your personal folder

8. EXPECTED OUTCOME AND OBSERVATIONS

All widgets display live data with no placeholder values; the Region filter updates every other widget consistently; the trend line renders clearly across the full period; and the layout follows a clear visual hierarchy with no overlaps.

Check	Observed (tick after running)
Dashboard connected to the RetailPulse Customers data module	[]
Summary card shows a real value for total active customers	[]
Line chart shows New and Returning lines across the full period	[]
Bar chart shows one bar per loyalty segment	[]
Selecting a Region updates the card, line chart and bar chart	[]
Clearing the filter returns all widgets to the full data	[]
No widgets overlap and the layout reads top to bottom	[]
Dashboard saved	[]

9. CONCLUSION

In this practical a customer insights dashboard was built in IBM Cognos Analytics from the RetailPulse Customers data module. The summary card gave the headline number of active customers, the line chart showed the trend of new and returning customers, and the bar chart showed the loyalty segment mix. A shared Region filter linked the widgets so that one selection updated the whole canvas, and an ordered layout made the dashboard easy to read. The practical showed how several visualizations can be combined into one interactive view for customer analysis.

10. PRECAUTIONS

- Use one data module for all widgets so the Region filter can apply to each of them.
- Set the filter scope to all widgets rather than a single widget.
- Check that the date item uses a sensible level, such as month, so the trend line is readable.
- Use clear widget titles and avoid overlap when resizing widgets.
- Save the dashboard after testing the filter.